

ATC-P6: Assistive Braking Device for Biking

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Presentation plan

- Definition of User Requirements
- Definition of System Specifications
- Existing solutions
- Brainstorming: presentation of possible ideas
- Presentation of our concept
- Budget Plan
- Future planning
- Expected Challenges



What is the challenge?

- User with **ectrodactyly**: rare genetic condition causing **missing or malformed fingers**
- A world not shaped for them: **issues** with **grasping** and **holding** objects
- Their wish: **biking safely**

What are the needs of our challenger?

Safety

- ❑ **Secure grip** for stable maneuvering
- ❑ **Ability to brake** with the left hand
- ❑ **Modulable braking control**
- ❑ **Immediate hand release** in emergencies
- ❑ Mechanism that avoids accidental braking activation

Comfort

- ❑ Not worn by the challenger, but **mounted on the bike**
- ❑ **Natural, coordinated hand movement**
- ❑ Ergonomic device, shaped for the user

Portability & Compatibility

- ❑ Easily mountable **without assistance**
- ❑ **Compatible with different bike types**
- ❑ Portable and easy to carry

Designing a system that matches those needs

Ergonomics

- **Custom-shaped grip** for secure hold and quick release
- Natural movement for braking activation: **wrist push**
- **Symmetrical** interaction for **both hands**

Mounting & Adaptability

- Opening/closing mechanism for quick attachment: **BOA Fit System**
- Compatible with **different handlebar diameters** and brake geometries

Portability

- **Lightweight** design
- Easy to install independently

Braking system

- Direct mechanical actuation of the brake
- **Progressive braking** proportional to applied force
- **Passive system**
- Integrated spring mechanism to prevent accidental activation

Review of existing solutions

1. The tool the user already has:



Front view of the user's current tool



Back view of the user's current tool

Review of existing solutions

1. The tool the user already has:



bike extension



Review of existing solutions

1. The tool the user already has:

Safety

- ✓ Secure grip
- ✗ Ability to brake
- ✗ Modulable braking control
- ✗ Immediate hand release



Review of existing solutions

1. The tool the user already has:

Comfort

- ✗ Not worn by the challenger
- ✗ Natural hand movement
- ✗ Non-bulky, ergonomic device



Review of existing solutions

1. The tool the user already has:

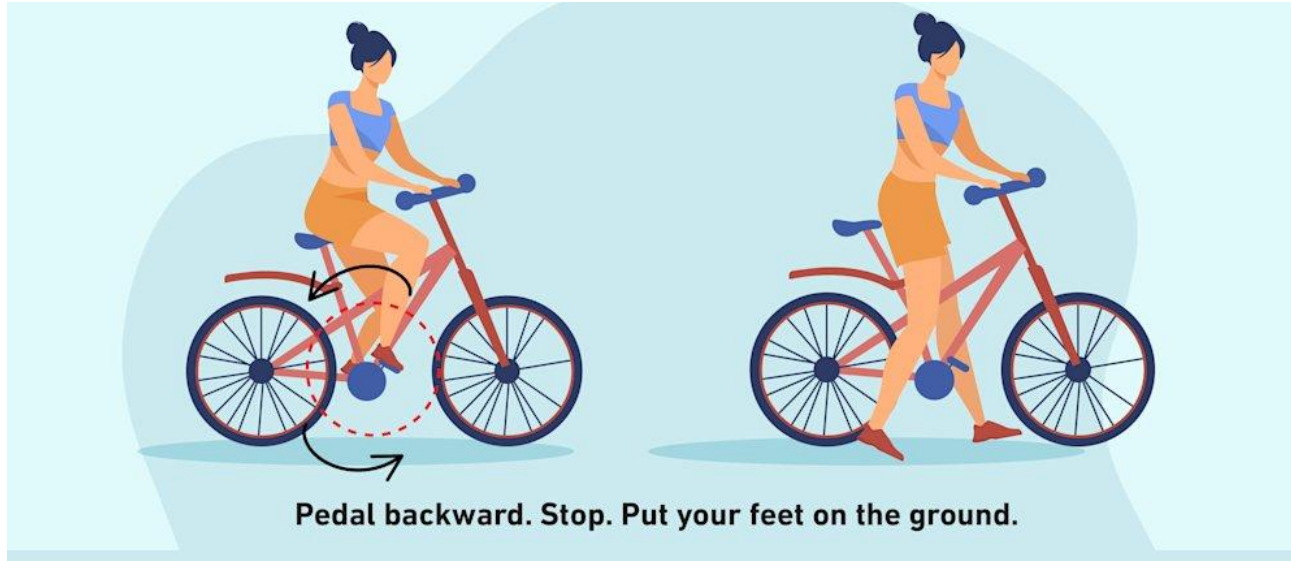
Portability & Compatibility

- ✓ Mountable without assistance
- ✗ Compatible
- ✓ Portable



Review of existing solutions

2. Coaster Brake (pedal braking):



Review of existing solutions

2. Coaster Brake (pedal braking):

Safety

- ✗ Secure grip
- ✓ Ability to brake
- ✓ Modulable braking control
- ✓ Immediate hand release



Review of existing solutions

2. Coaster Brake (pedal braking):

Comfort

- ✓ Not worn by the challenger
- ✗ Natural movement
- ✓ Non-bulky, ergonomic device



Review of existing solutions

2. Coaster Brake (pedal braking):

Portability & Compatibility

- ✗ Mountable without assistance
- ✗ Compatible
- ✗ Portable



Review of existing solutions

3. Project "Caroline" by Macu4:



Project "Caroline" custom design installed on real bike

Review of existing solutions

3. Project "Caroline" by Macu4:

Safety

- ✗ Secure grip
- ✓ Ability to brake
- ✓ Modulable braking control
- ✓ Immediate hand release



Project "Caroline" custom design installed on real bike

Review of existing solutions

3. Project "Caroline" by Macu4:

Comfort

- ✓ Not worn by the challenger
- ✓ Natural hand movements
- ✓ Non-bulky, ergonomic device



Project "Caroline" custom design installed on real bike

Review of existing solutions

3. Project "Caroline" by Macu4:

Portability & Compatibility

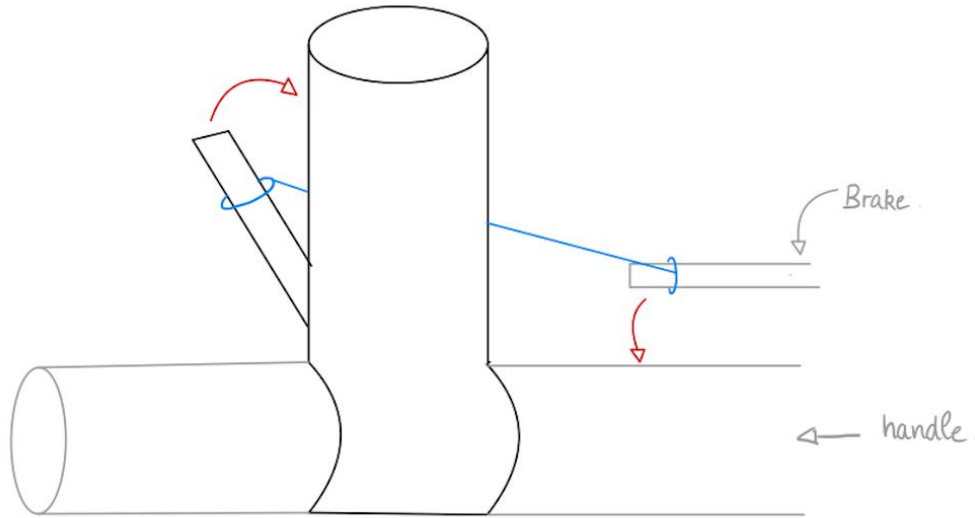
- ✗ Mountable without assistance
- ✗ Compatible
- ✗ Portable



Project "Caroline" custom design installed on real bike

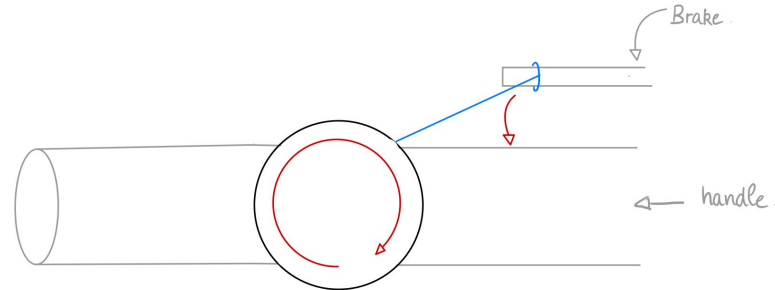
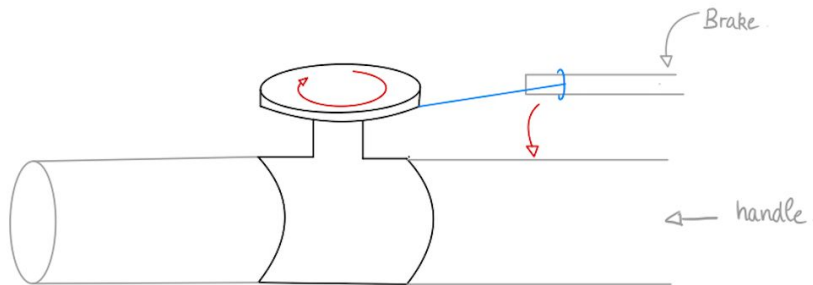
Concept brainstorming:

1. Primal ideas



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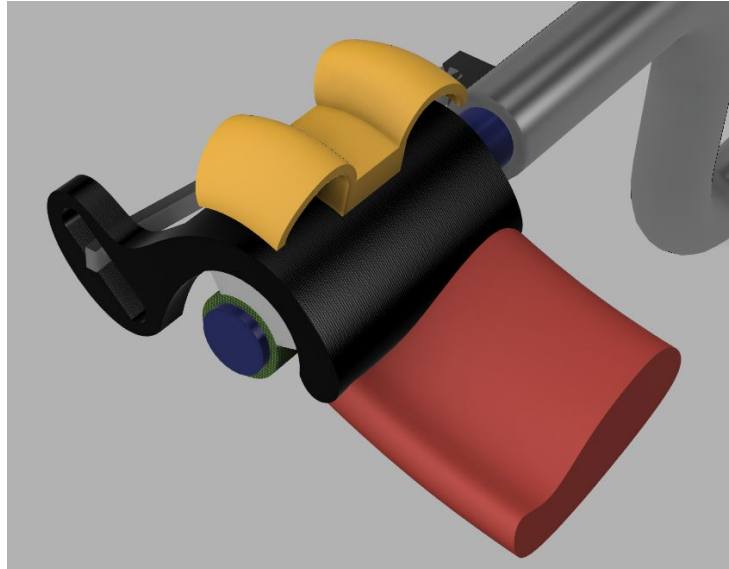
2. After the feedback

⇒ **Adapt the Macu4 mechanism**



Idea approved by:

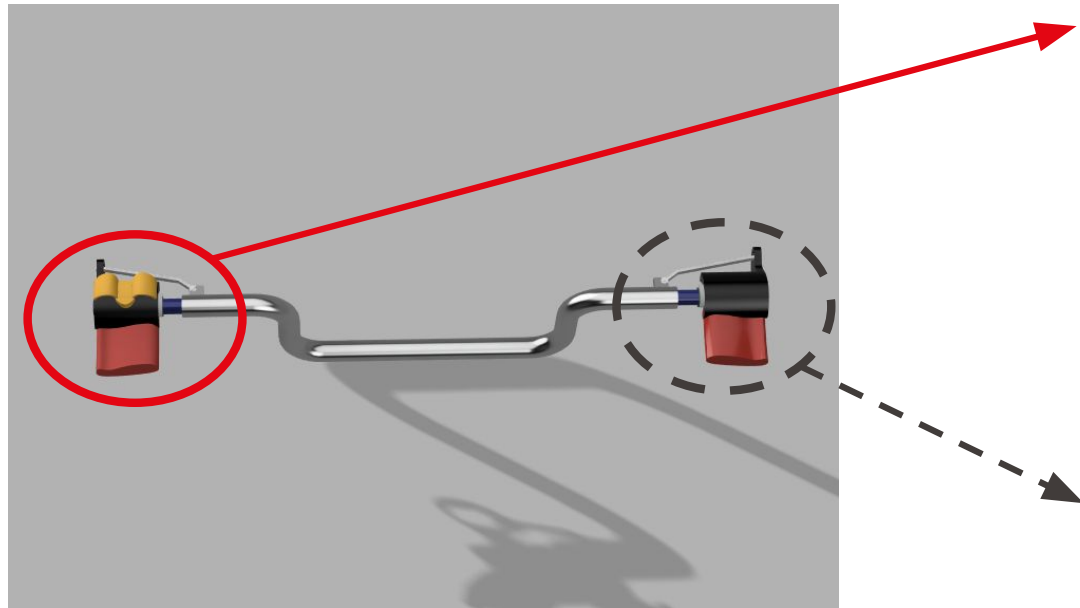
- ✓ The challenger
- ✓ The challenger's physical therapist
- ✓ The founder of Macu4



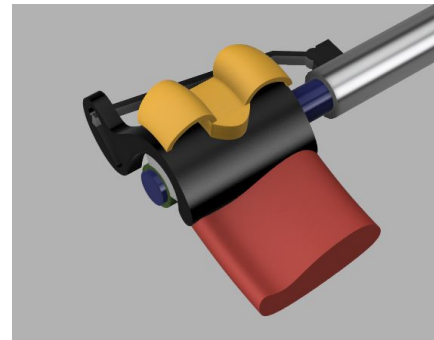
Isometric view of the concept

Design of the prototype

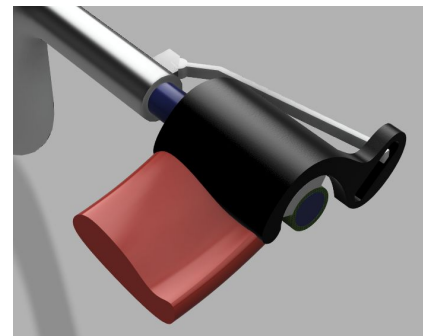
The concept



Overview of the concept

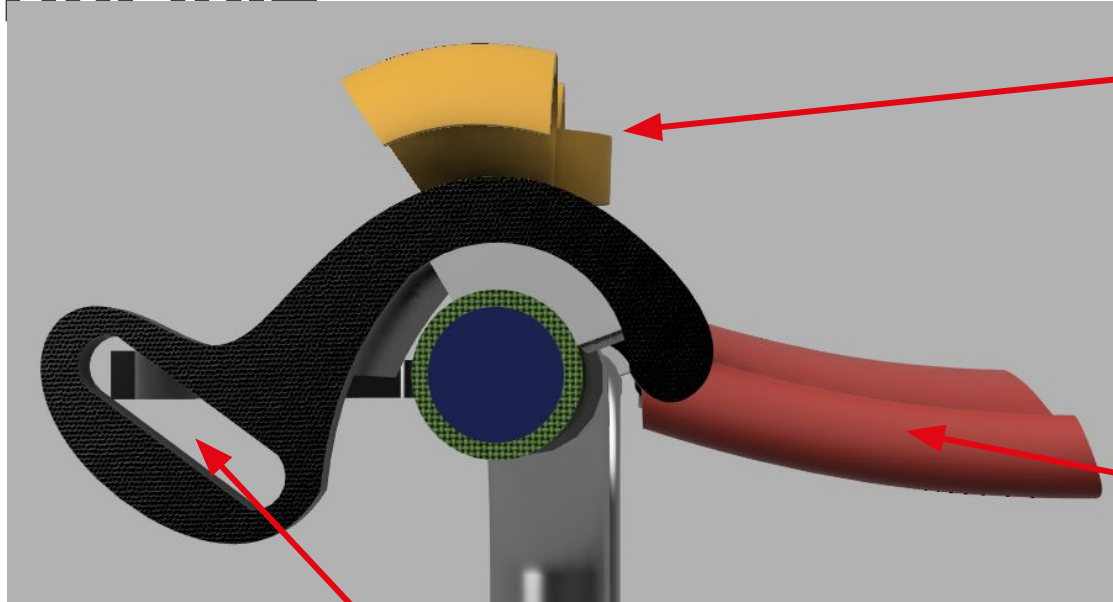


Left-hand assistive device
(non-functional hand)



Right-hand device
(functional hand)

Working principle

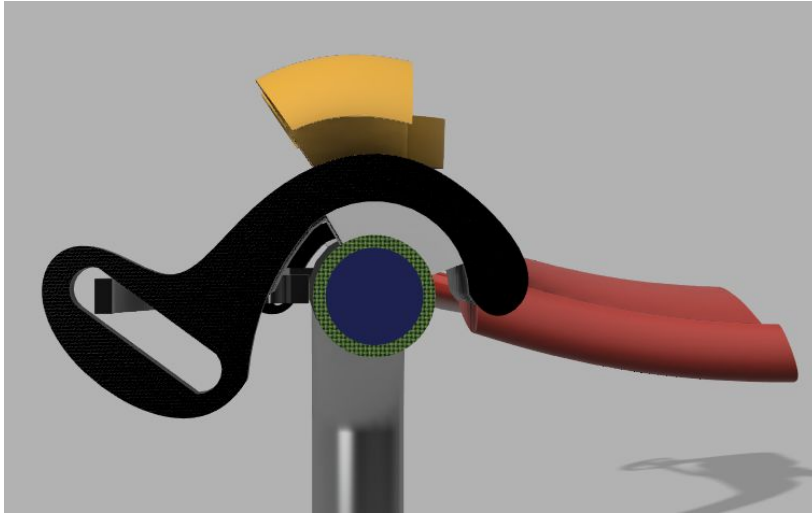


The fingers of the user will fit here

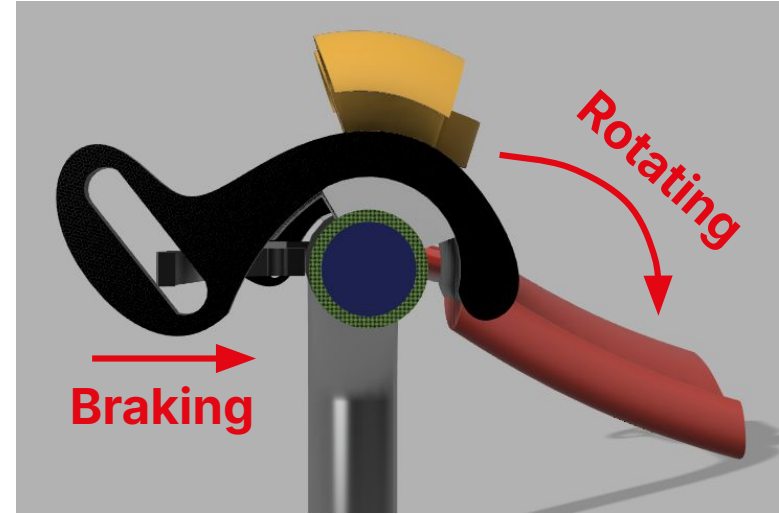
The palm of the user will rest here

Mechanism acting on the brakes

Working principle

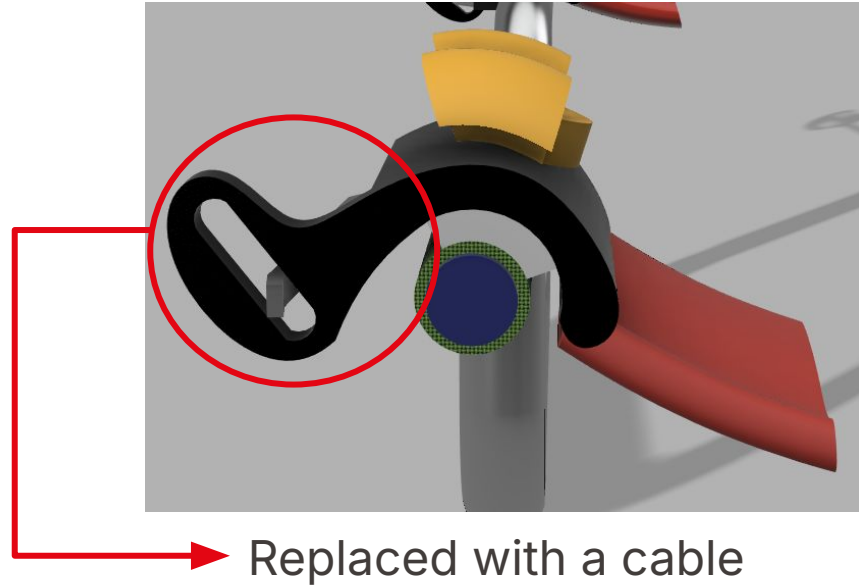
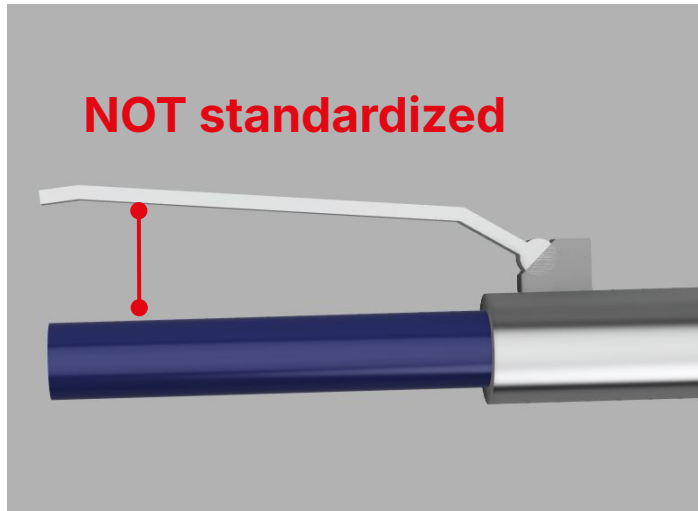


Resting position



Brakes activation

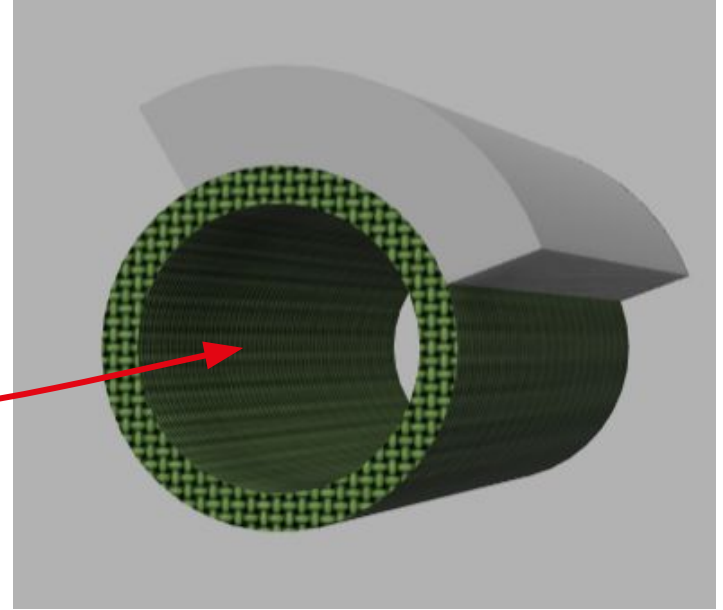
Adaptability on any bike



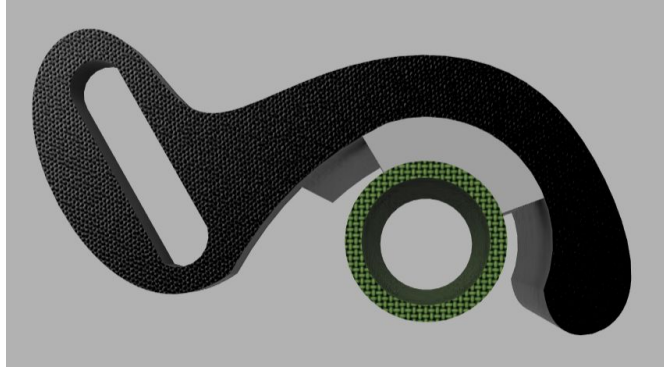
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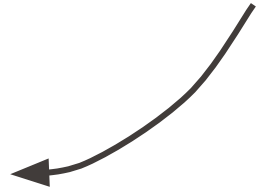
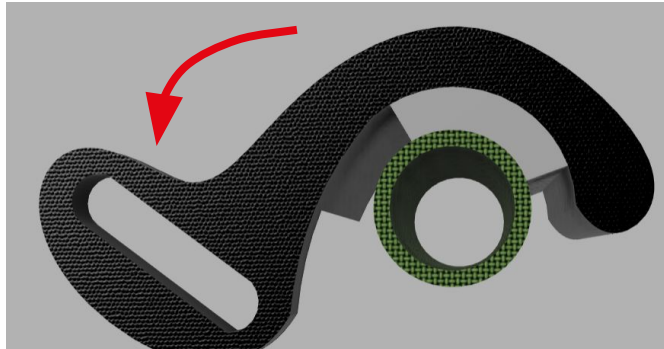
BOA Fit System



Pivot mechanism

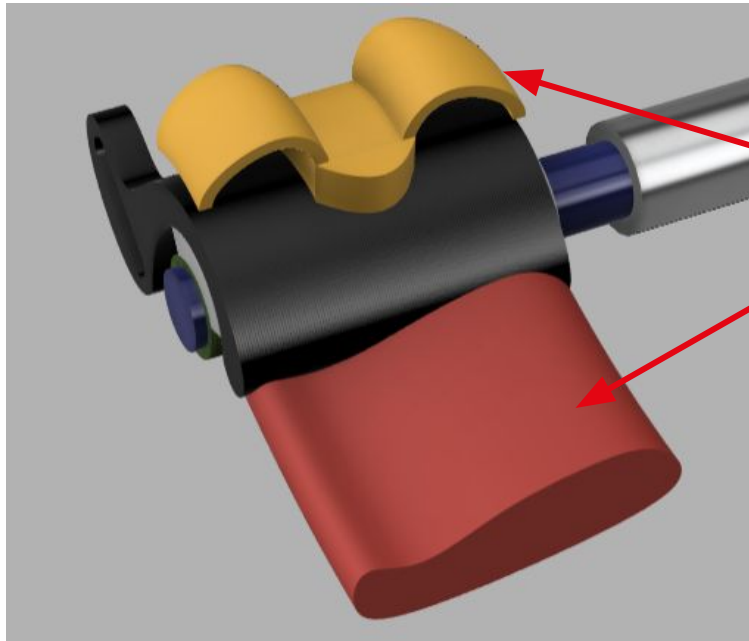


Frame can rotate
relative to the grey
slider block



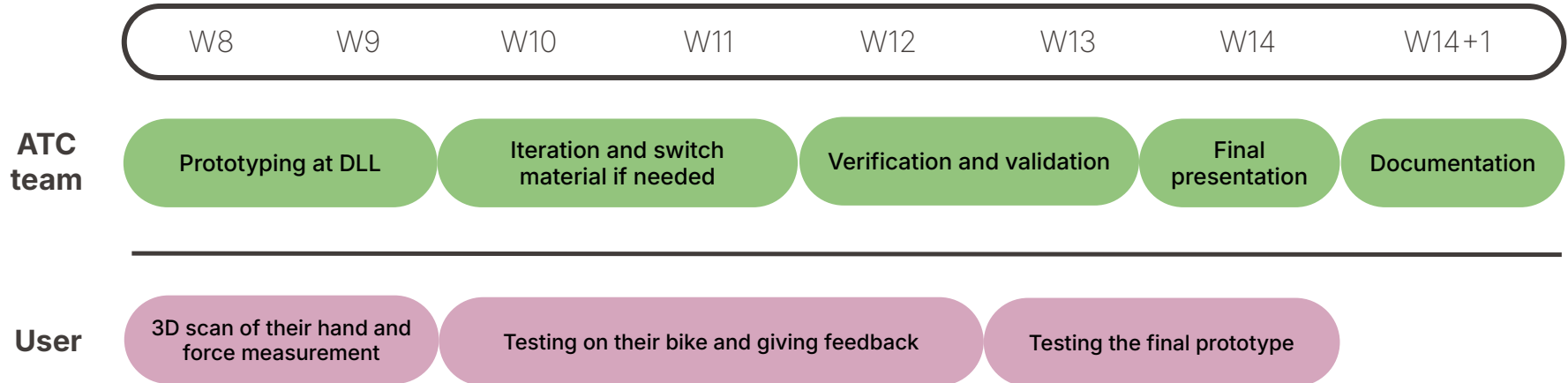
Dovetail sliding
mechanism (in the
grey block)

Comfort (3D molding)



Adapted to the user,
3D molding

Planning



Budget Plan

Category	Amount (CHF)	% Total	Notes
Travel & visits	50.-	5.8%	
PLA 3D printing	150.-	17.6%	For testing & prototyping in 3D
Mechanical components	50.-	5.8%	Ball bearing, screws, springs, sliders...
BOA system	60.-	7%	Or similar
Stainless steel cable	40.-	4.7%	Sertissage & cable
Metal and composite machining	500.-	58.8%	PLA probably not strong enough
Total	850.-	100%	Initial allocated budget (1000.-)

Risk Assessment

ID	Hazard	Risk	S	O	R	Mitigation Measure
R1	Mechanical failure	Device does not brake/slips	3	2	6	Material testing, safety factor in design, user testing protocol
R2	Accidental activation	Brake triggered unintentionally while cycling	4	2	8	Lo-ratchet safety mechanism; resistance threshold tested with user
R3	Poor fit/comfort	Device is painful or unusable by challenger	3	2	6	Iterative prototype fitting sessions; adjustable straps & sizing
R4	Bike incompatibility	Device cannot attach to all bike models	2	3	6	Universal mount design; tested on 2+ bike types before delivery

Any question?